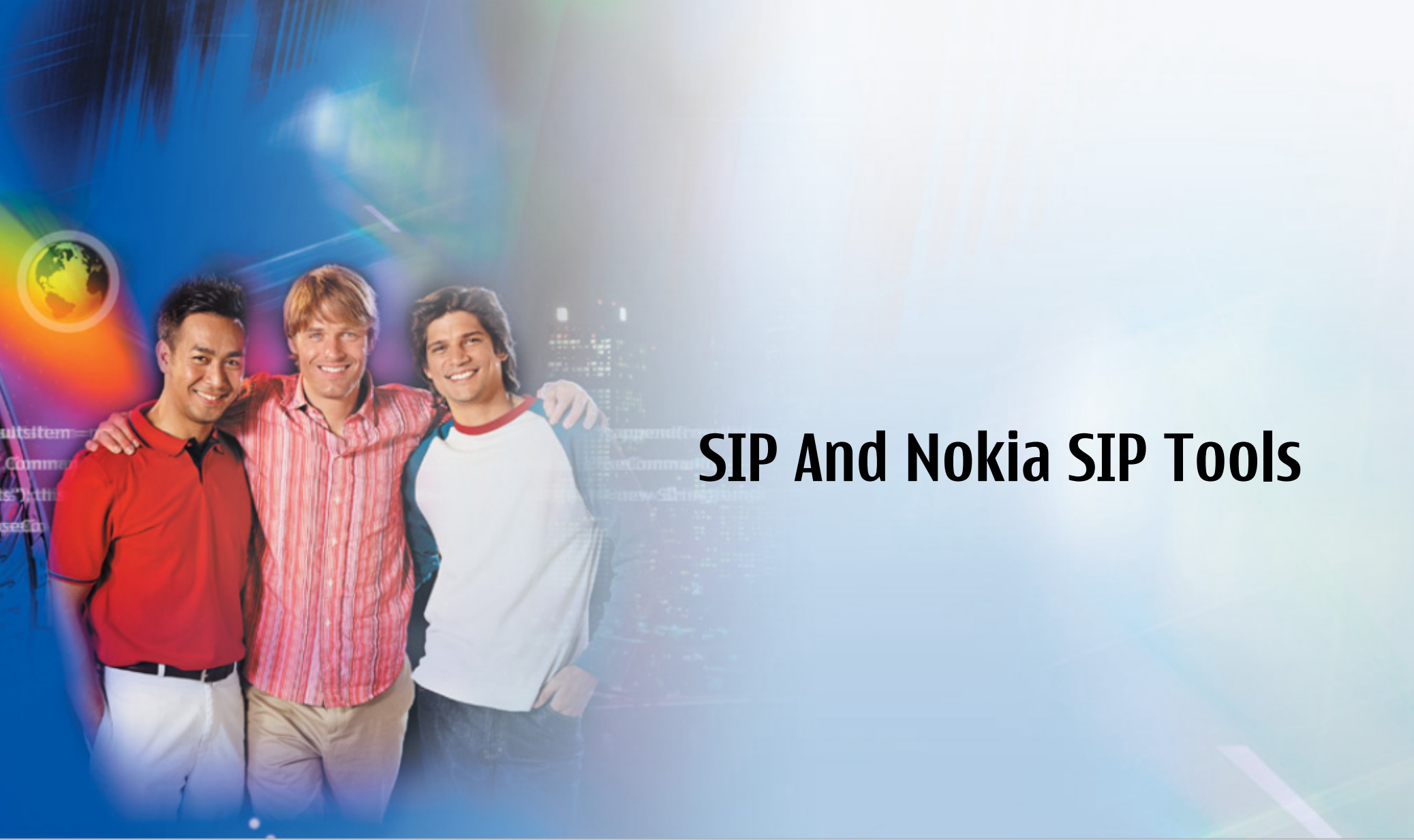


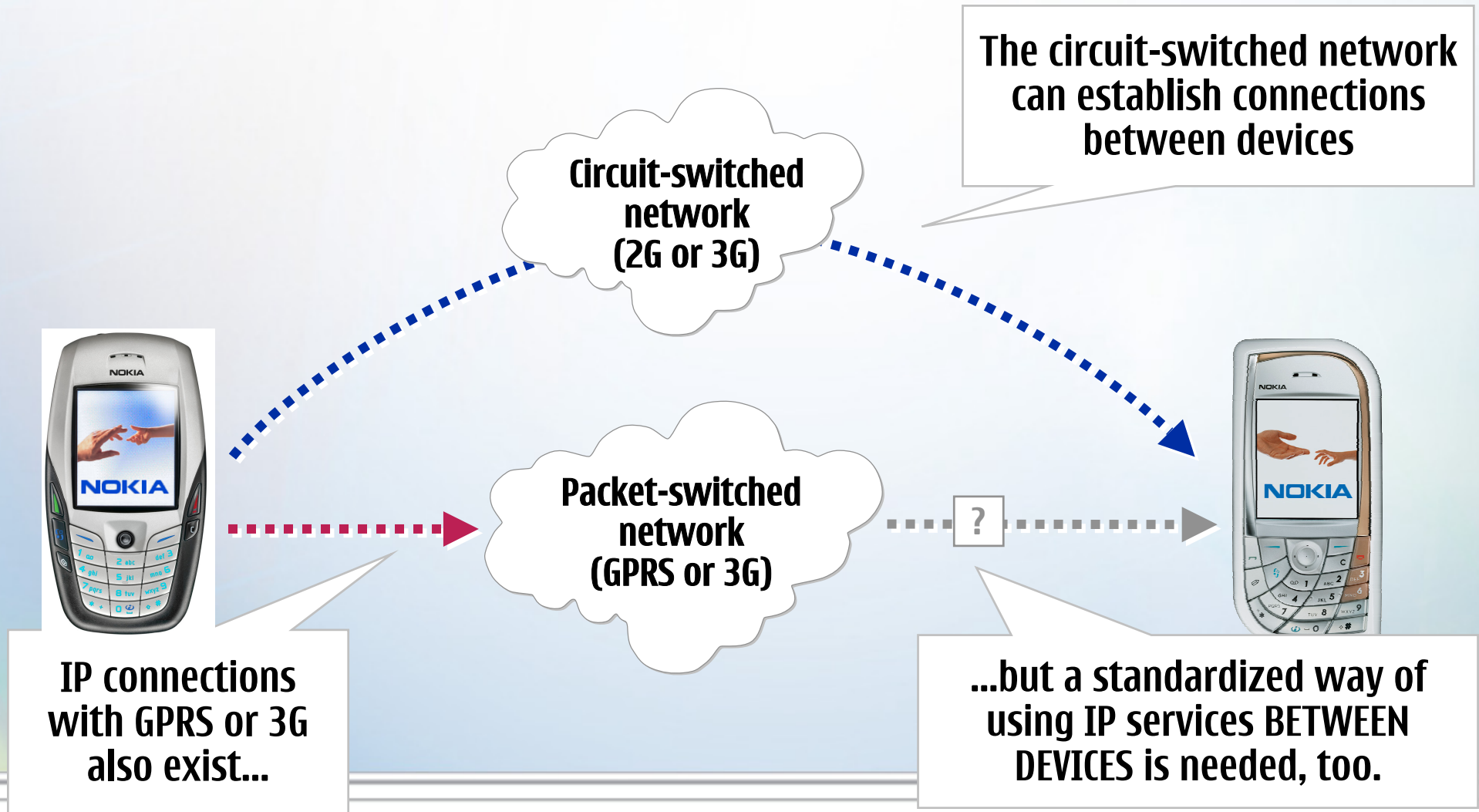


SIP And Nokia SIP Tools

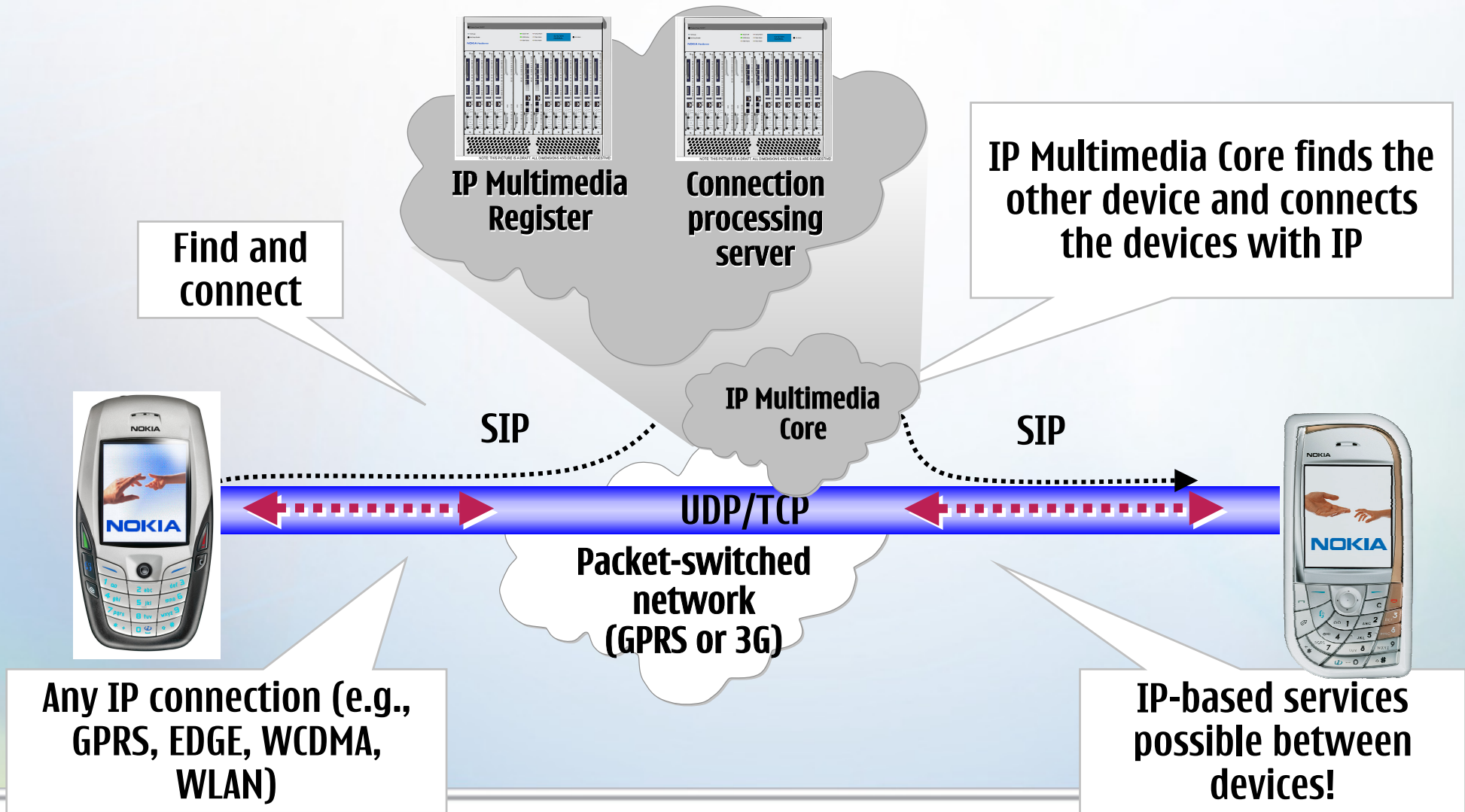
Content

- **Person-to-person IP Connections**
- **What is SIP?**
- **SIP as a Service Enabler**
- **Nokia Tools and Support for SIP**

Connecting People for Multimedia?



Person-to-person IP Connection



What Is SIP?

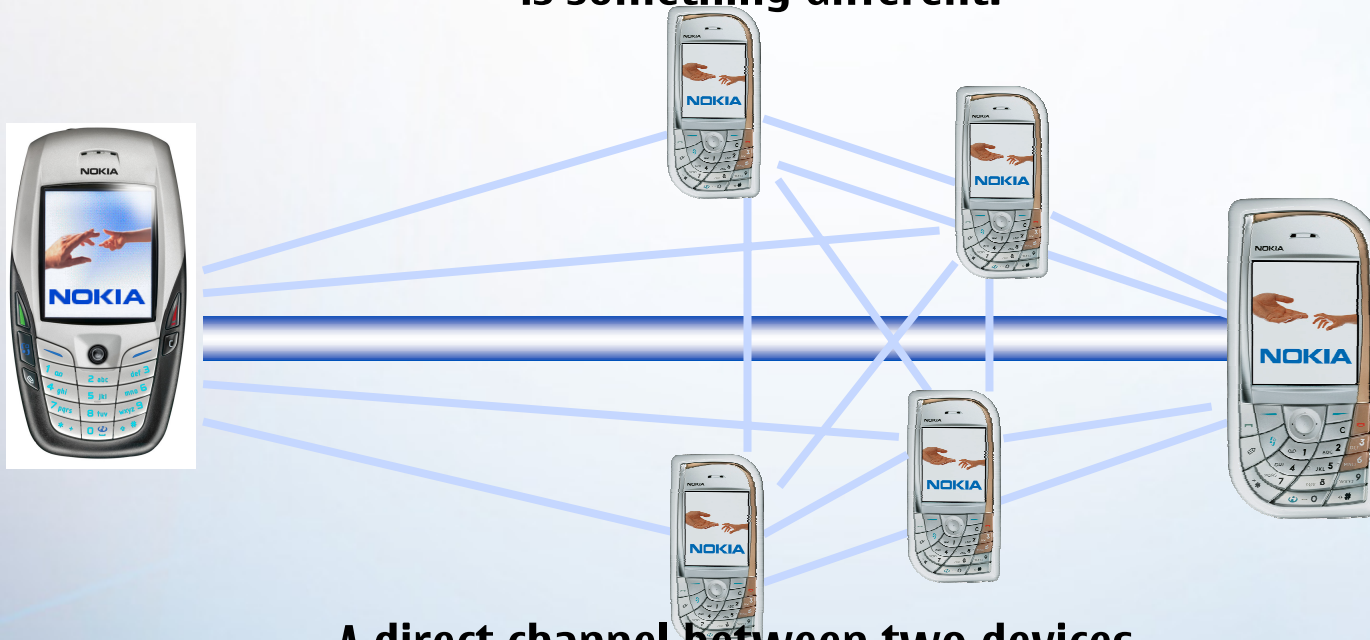
- **Session Initiation Protocol**
 - Defined by the Internet Engineering Task Force (IETF)
 - Designed after HTTP
- SIP is used for initiation of interactive communications sessions between users. SIP also handles termination and modifications of sessions as well.
- Although originally designed for multimedia sessions
 - Scope has been extended: presence, gaming, IM, etc.
 - SIMPLE: SIP for Instant Messaging and Presence Leveraging Extensions
- Smarter than your usual transport
 - Personal addressing, routing, etc.
- Since its approval in early 1999 as an official standard, SIP has gained tremendous market acceptance for signaling communications service on the Internet.
- Adopted by 3GPP in Release 5
 - All mobile operators know that they will be operating SIP infrastructure
- More information from IETF RFC3261

Benefits of SIP

- SIP is a text-based language and very similar to HTTP. It is easy to program compared to legacy mobile network protocols.
- SIP is gaining acceptance in the fixed Internet as the protocol for session establishment. Competence developed in the Internet community can be utilized in the mobile domain.
- Use of IP and SIP enables the convergence of fixed and mobile Internet domains.
- SIP can be used to implement all types of peer-to-peer applications, e.g.,
 - Voice
 - Video
 - Messaging
 - Gaming
 - Push to talk

SIP as a Service Enabler

**What we can offer in an IP-based world
is something different:**



**A direct channel between two devices
or a group of devices...
and applications running in the devices.**

Use Cases for IP Multimedia

Peer-to-peer connectivity can be used for various purposes:

- Sharing data and media between devices (gaming, pictures, music, contacts, PoC, etc.)
- Contacting other devices (PDAs, PCs, cams) and using the data stored there



Interactive Applications – by which Tools?

- Standard toolkit for application developers
- Standard Nokia SIP component for mobile devices
- Standard programming interface (API) for the SIP stack

Nokia SIP plug-in for Series 60 SDK

Nokia SIP Plug-in for Series 60 SDK Enables Services

Enables:

- ***Lower threshold*** to develop SIP-enabled applications – SIP API hides the network functionality from the developed application.
- ***Fast development times*** – Industry-leading development tools meaning that developers do not have to be SIP experts.
- ***Fast deployment*** – Connected services that do not require new servers can be deployed quickly.
- ***Leveraging Nokia's mobile penetration*** – Developed applications have potential multi-million size market. Starting with Series 60 (Symbian 7.0s based) and expanding to other Nokia platforms.
- ***Less fragmentation*** – Emergence of different implementations of SIP could potentially cause fragmentation of services.
- Better interoperability of SIP services.

Nokia SIP Plug-in for Series 60 SDK Enables Services

How:

- **Piloting of the Nokia SIP plug-in SDK and selected developer applications with major mobile operators.**
- **Network infrastructure to enable these new services has been commercially introduced during 2004.**
- **Services can be used in existing 2.5 G networks and all future networks.**
- **Both Nokia network and device business are committed to introducing IMS system.**
- **Many operators are currently running pilots to prepare for commercial launch of IMS services.**

Nokia Tools and Support

- **SIP plug-in for Series 60 SDK** provides tools and documentation for developing and testing SIP-enabled applications
 - SIP extensions for Series 60 2.1 device emulator and device
 - SIP stack and API
 - SIP server emulator
 - Example application
 - User's Guide and Programmer's Guide
- **Nokia SIP Component**
 - Includes SIP stack and SIP settings UI for distribution with SIP applications
- Tools are available for download from the Forum Nokia Web site (<http://www.forum.nokia.com>).
- Standard Forum Nokia developer support for the tools is available.

Supported Devices

- The Nokia SIP component for Nokia devices is available
 - For development use with the SIP plug-in for Series 60 SDK
 - For distribution with SIP applications for Nokia Series 60 devices (note, requires acceptance of distribution agreement before download)
- The Nokia SIP component is supported in the following Nokia Series 60 devices:
 - Nokia 6600 device
 - Nokia 7610 device
 - Nokia 6260 device
- The Nokia 6260 device includes the Nokia SIP stack as pre-installed software together with the Push to talk application. However, the SIP settings are not supported. Therefore the Nokia SIP component must be installed to the device to add the SIP settings.

Series 60 SIP Development Environment

- Nokia Series 60 SDK 2.1 for Symbian OS has to be installed before installing the SIP plug-in.
- The whole environment with two Series 60 device emulators and SIP Server Emulator can be installed and run on one PC.
- Testing in the emulator environment can be done by using SIP Server Emulator to emulate SIP services of the communication network.
- Provided SIP Server Emulator implements SIP Proxy and Registrar Server functionality.

